PROJECT PHASE 2

Statistical Damage Detection

Feature Engineering & Anomaly Detection

Course:	Sensing Techniques and Data Analytics
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Institution:	University of Alberta
Due Date:	Thursday, November 19, 2026 at 11 AM (before class)
Weight:	20% of final grade (100 points)

1 PROJECT NARRATIVE

Project Background

Following Phase 1's modal analysis of the Highway 43 bridge, your team must now develop automated statistical detection methods to monitor future structural changes without manual inspection of every measurement. Your colleagues at the structural health monitoring firm have processed the FDD analysis results and compiled them into a standardized JSON file (`phase2_modal_parameters.json`). This file contains natural frequencies and mode shapes for three structural conditions: intact (baseline), damage scenario 1, and damage scenario 2.

Environmental effects such as temperature variations and traffic loads cause natural fluctuations in modal parameters even in healthy structures. You will implement three statistical methods—z-score for rapid screening, Mahalanobis distance for correlation-aware detection, and Wasserstein distance for distribution comparison—progressing from single-feature analysis through bivariate methods to high-dimensional feature fusion with PCA.

Learning Objectives:

- Understand how statistical distributions reveal structural condition changes
- Apply univariate detection methods (z-score, Wasserstein distance)
- Implement multivariate techniques that account for feature correlation
- Use dimensionality reduction (PCA) to extract dominant damage patterns
- Compare detection performance across different feature combinations

2 DATA STRUCTURE

Provided Data File

Your colleagues have prepared `phase2_modal_parameters.json`, which contains modal parameters extracted from ambient vibration tests and FDD. This file includes results from 10 independent measurements, with each measurement divided into two time segments to provide more statistical samples.

```
{
  "scenarios": {
    "intact": {
      "natural_frequencies": {
        "mode_1": [20 values from 20 measurements],
        "mode_2": [20 values],
        "mode_3": [20 values]
      },
      "mode_shapes": {
        "mode_1": {
          "measurement_1": [24 values at sensor locations],
          ...
          "measurement_20": [24 values]
        },
        "mode_2": {...},
        "mode_3": {...}
      }
    },
    "damage1": {...same structure...},
    "damage2": {...same structure...}
  },
  "metadata": {
    "sampling_info": {
      "original_measurements": 10,
      "segments_per_measurement": 2,
      "total_estimates": 20
    },
    "sensor_assignments": {
      "group_1": ["S1", "S2", "S5", "S7", "S9", "S13", "S16", "S19", "S22"],
      ...
    }
  }
}
```

Understanding the Data:

- 10 ambient vibration tests were conducted, each recorded for 60 seconds
- Each recording was divided into two 30-second segments for statistical sampling
- This provides 20 independent estimates per structural condition
- Three natural frequencies are included (one per mode)
- Mode shapes are provided at all 24 sensor locations on the bridge (3 mode shapes)
- Use **ONLY** the 9 sensors assigned to your group for mode shape calculations (Refer to Phase 1)
- Each mode shape is a list of 24 values ordered S1 to S24, so sensor Sk is at list index $k-1$ (see the starter code)

3 ANALYSIS TASKS

3.1 Part A: Feature Distribution Analysis *25 points*

A1. Natural Frequency Distributions (8 pts)

- For each mode (1, 2, 3), create one figure with three subplots showing frequency histograms for intact, damage1, and damage2 scenarios
- Fit Gaussian distributions to each histogram using maximum likelihood estimation
- Overlay fitted probability density functions and report mean (μ) and standard deviation (σ)
- Add vertical lines at means and shaded $\pm\sigma$ regions for visual clarity
- Produce three figures total (one per mode)
- Discuss: How do frequency distributions shift across damage scenarios? Which mode shows the strongest sensitivity?

A2. Mode Shape Visualization (9 pts)

- Create one comprehensive figure with a 3×3 subplot layout (3 modes × 3 scenarios)
- Each column represents one structural condition: intact (left), damage1 (middle), damage2 (right)
- Each row represents one mode: mode 1 (top), mode 2 (middle), mode 3 (bottom)
- For each subplot, plot the mean mode shape across all 20 measurements as a solid line (as a function of sensor position)
- Add a shaded region showing mean ± 1 standard deviation to visualize uncertainty (STD is different for each sensor position)
- X-axis: sensor positions along the bridge (use your 9 assigned sensors)
- Y-axis: normalized mode shape amplitude
- Include subplot titles indicating mode number and scenario
- Discuss: How does mode shape uncertainty (standard deviation) change with damage? Are damaged regions showing increased variability?

A3. Modal Assurance Criterion (MAC) Distributions (8 pts)

- Compute the mean mode shape vector from all 20 intact measurements for each mode (3 mode shapes)—this serves as your reference
- For each mode, compute MAC values comparing all the 20 estimates from each condition (intact, damage1, damage2) to the corresponding reference mean mode shape
- Create three figures (one per mode), each with three subplots showing MAC histograms for all scenarios
- Fit Gaussian distributions and report μ and σ for each
- Discuss: How do MAC distributions change with damage? Which mode shows the clearest separation between conditions?

Modal Assurance Criterion (MAC)

The MAC quantifies the similarity/correlation between two mode shapes. For a reference shape ϕ_{ref} (mean of the 20 intact estimates) and a test shape ϕ_{test}:

$$\text{MAC} = \frac{|\phi_{ref}^T \phi_{test}|^2}{(\phi_{ref}^T \phi_{ref})(\phi_{test}^T \phi_{test})}$$

A MAC value of 1.0 indicates identical shapes, while values below 0.9 suggest significant differences. When computing MAC, use only the mode shape values at your 9 assigned sensor positions. The reference shape should be the mean of all 20 intact measurements for robust comparison.

3.2 Part B: Single Feature Damage Detection *20 points*

Analyze all six features individually: $f_1, f_2, f_3, \text{MAC}_1, \text{MAC}_2, \text{MAC}_3$

B1. Z-Score Anomaly Detection (10 pts)

- For each feature, use intact measurements (n=20) as baseline: compute μ_0, σ_0
- For each damaged measurement, calculate how many standard deviations it differs from the intact mean:

$$z = \frac{x - \mu_0}{\sigma_0}$$

- Flag observations with $|z| > 2$ as anomalies (95% confidence level)
- Calculate detection rate: percentage of damaged measurements correctly flagged
- Create detection summary table showing baseline statistics and detection rates for both damage scenarios

B2. Wasserstein Distance Analysis (7 pts)

- For each feature, compute the Wasserstein distance (Earth Mover's Distance) between intact and damage distributions
- Normalize by dividing by the intact standard deviation: $W_1^{norm} = W_1/\sigma_0$
- This allows fair comparison across features with different units and scales
- Report normalized distances in a table format
- Create bar chart comparing W_1^{norm} across all 6 features for both damage scenarios. The x-axis should show the six feature names with grouped bars for each damage scenario.

B3. Feature Sensitivity Ranking (3 pts)

- Rank features by detection performance (use Damage2 results for primary ranking)
- Identify which single feature is most damage-sensitive and which is least sensitive
- Compare frequency-based features versus MAC-based features: which type performs better overall?

3.3 Part C: Bivariate Analysis *20 points*

Analyze two feature pairs: (f_1, f_2) and $(\text{MAC}_1, \text{MAC}_2)$

C1. Joint Distribution Visualization (7 pts)

- For each feature pair, create one figure with three subplots (1×3 layout)
- Each subplot shows scatter plot with fitted bivariate Gaussian contours for one scenario
- Overlay 95% confidence ellipse: $(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) = \chi^2_{0.95}(2) \approx 5.99$
- Compute and display correlation coefficient ρ on each subplot
- Produce two figures total (one per feature pair)
- Discuss: Are (f_1, f_2) correlated? What about $(\text{MAC}_1, \text{MAC}_2)$? Explain physically why or why not.

C2. Mahalanobis Distance Detection (8 pts)

- For each feature pair and both damage scenarios, compute Mahalanobis distances using the intact baseline:

$$D_M = \sqrt{(\mathbf{x} - \boldsymbol{\mu}_0)^T \boldsymbol{\Sigma}_0^{-1} (\mathbf{x} - \boldsymbol{\mu}_0)}$$

- This metric accounts for correlations between features, unlike independent z-scores
- Apply threshold: $D_M > \sqrt{\chi^2_{0.95}(2)} \approx 2.45$ to identify anomalies
- Create a 2×2 comparison figure showing scatter plots with detection boundaries and D_M histograms
- Report detection rates in a clear table

C3. Bivariate vs Univariate Comparison (5 pts)

- Compare Mahalanobis (bivariate) detection rates with z-score (univariate) rates from Part B
- Create a comparison table showing detection performance for individual features versus combined corresponding features
- Does accounting for correlation improve detection?
- Between frequency pair versus MAC pair: which is more effective for damage detection?

3.4 Part D: Feature Fusion and Engineering *15 points*

Explore different 4-feature combinations from the 6 available features.

D1. Systematic Combination Testing (10 pts)

- Test all possible ways to select 4 features from the 6 available: $\binom{6}{4} = 15$ unique combinations
- For each combination, build 4D feature vectors for all measurements
- Compute multivariate Wasserstein distance between intact and each damage scenario
- Create two bar charts (subplots) in the same figure (one per damage scenario) comparing all combinations
- Label x-axis with combination identifiers (e.g., "$f_1, f_2, f_3, \text{MAC}_1$")
- Identify the top 3 most sensitive combinations for each damage type

D2. Feature Importance Analysis (5 pts)

- Analyze which features appear most frequently in top-performing combinations
- Compare frequency-dominated versus MAC-dominated combinations: which type is more sensitive?
- Compare the best 4-feature combination with the best single feature from Part B
- Quantify the improvement: does feature fusion significantly enhance detection sensitivity?
- Discuss whether features provide independent information or redundant insights

3.5 Part E: Principal Component Analysis *15 points*

Systematically analyze dimensionality reduction from 6D to lower dimensions using PCA.

E1. PCA Implementation and Variance Analysis (4 pts)

- Standardize all 6 features of all scenarios to zero mean and unit variance using intact baseline statistics (z-score normalization)
- Fit PCA on the intact data only (n=20)
- Create scree plot showing variance explained by each principal component
- Report: How many PCs are needed to capture 95% of the variance?

E2. First Principal Component Analysis (3 pts)

- Transform all the damage measurements to PC1 space
- Create one figure with overlaid distributions for all three scenarios with fitted PDFs
- Compute Wasserstein distance $W_1(\text{intact}, \text{damage}_k)$ for PC1
- Compare with Part B results: is PC1 more sensitive than any single feature?

E3. Two-Dimensional PCA Space (3 pts)

- Create 2D scatter plot: PC1 versus PC2, color-coded by scenario (intact, damage1, damage2)
- Overlay 95% confidence ellipses for each scenario
- Verify that PC1 and PC2 are uncorrelated for intact data (fundamental PCA property)
- Compute 2D Wasserstein distance in PC1-PC2 space
- Compare with PC1-only result: does adding PC2 improve detection sensitivity?

E4. Progressive Dimensionality Impact (3 pts)

- Compute Wasserstein distances using different numbers of components: $k = 1, 2, 3, 4, 5, 6$
- Create one figure with two subplots (one per damage scenario) showing distance versus number of PCs
- Identify optimal k: where does detection sensitivity plateau?
- Compare 6-PC result with full 6D features (no PCA): are they identical? Why?

E5. Physical Interpretation (2 pts)

- Report PC1 loadings: which original features contribute most to PC1?
- Does PC1 represent a physically meaningful quantity (e.g., "overall structural stiffness")?
- Explain why dimensionality reduction works for this problem: what variance is being discarded?

3.6 Part F: Engineering Insights *5 points*

Provide a comprehensive discussion addressing these questions:

F1. Method Comparison and Selection (2 pts)

- Rank all detection approaches by effectiveness: single features, bivariate, 4D fusion, PCA
- Which method would you deploy for a real-time monitoring system? Justify considering both accuracy and computational cost.

F2. Feature Selection Strategy (2 pts)

- If limited to monitoring only 2 features due to computational constraints, which would you choose? Why?
- Are mode shapes (MAC values) necessary, or do natural frequencies provide sufficient information?
- Propose a hierarchical detection system that uses progressive complexity (e.g., fast screening followed by detailed analysis)

F3. Practical Implementation Considerations (1 pt)

- How would seasonal temperature variations affect your baseline distribution?
- Suggest strategies to update baseline statistics over time without losing damage sensitivity
- Discuss false positive versus false negative trade-offs: how would you tune detection thresholds for different monitoring objectives?

4 DELIVERABLES

Submission Requirements

Submit **TWO files** via the course portal before the due date:
File 1: `ProjectPhase2_GroupX.zip`

```
ProjectPhase2_GroupX/
    Main.ipynb       # Complete notebook with all analysis
    figures/
    results/
    README.txt       # Brief description of contents
```

File 2: `ProjectPhase2_GroupX.pdf` - Exported notebook **with all outputs visible**

5 ACADEMIC INTEGRITY

While collaboration within groups is expected, each group must submit independent work. Code and interpretations must be original. Properly cite any external resources used.

Generative AI. AI tools are encouraged in this course, but you are responsible for every result you submit. Add a short note at the end of your notebook stating what you used AI for and your main prompts.

Marking. As stated in the syllabus, phases are marked comparatively: the strongest submission sets the standard for full marks, and the others are marked relative to it.

Late work. Late submissions lose 10% of this phase's value per day, up to three days, after which they are not accepted.

6 GRADING RUBRIC

Component	Points	Key Assessment Criteria
Part A: Distribution Analysis	25	Frequency/MAC distributions, mode shape visualization
Part B: Single Feature Detection	20	Z-score and Wasserstein implementation, sensitivity ranking
Part C: Bivariate Analysis	20	Mahalanobis distance, correlation analysis, comparisons
Part D: Feature Fusion	15	Combination testing, feature importance analysis
Part E: PCA	15	Variance analysis, dimensionality impact, interpretation
Part F: Engineering Insights	5	Practical recommendations, method comparison
Total	**100**	

7 IMPLEMENTATION GUIDELINES

Essential Implementation Requirements

- Extract mode shape values only from your 9 assigned sensor locations
- Label all axes clearly and include legends where multiple conditions are shown
- Include titles on all figures explaining what is being displayed
- For 1D Wasserstein distances use `scipy.stats.wasserstein_distance`; for the multivariate cases in Parts D and E use `scipy.stats.wasserstein_distance_nd` (SciPy 1.13 or newer)

```python
import json
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

# Load data
with open('phase2_modal_parameters.json', 'r') as f:
    data = json.load(f)

# Extract your assigned sensors
my_sensors = data['metadata']['sensor_assignments']['group_X']  # Replace X

# Extract frequencies for a scenario
def extract_frequencies(scenario, mode_key='mode_1'):
    return np.array(data['scenarios'][scenario]['natural_frequencies'][mode_key])

# Extract mode shape at assigned sensors
def extract_mode_shape(scenario, measurement_idx, mode_key='mode_1'):
    mode_data = data['scenarios'][scenario]['mode_shapes'][mode_key][f'measurement_{measurement_idx}']
    # mode_data is a list of 24 values ordered S1..S24, so S1 -> index 0
    return np.array([mode_data[int(s[1:]) - 1] for s in my_sensors])

# Calculate MAC between two mode shapes
def calculate_MAC(phi_ref, phi_test):
    numerator = np.abs(np.dot(phi_ref, phi_test))**2
    denominator = np.dot(phi_ref, phi_ref) * np.dot(phi_test, phi_test)
    return numerator / denominator

# Compute reference mode shapes (mean of all 20 intact measurements)
def compute_mean_mode_shape(scenario, mode_key):
    """Compute mean mode shape across all 20 measurements"""
    all_shapes = []
    for i in range(1, 21):
        shape = extract_mode_shape(scenario, i, mode_key)
        all_shapes.append(shape)
    return np.mean(all_shapes, axis=0)

# Get reference mode shapes (mean of intact measurements)
ref_mode1 = compute_mean_mode_shape('intact', 'mode_1')
ref_mode2 = compute_mean_mode_shape('intact', 'mode_2')
ref_mode3 = compute_mean_mode_shape('intact', 'mode_3')

# Mahalanobis distance calculation
def mahalanobis_dist(x, mu, cov):
    cov_inv = np.linalg.inv(cov)
    diff = x - mu
    return np.sqrt(diff @ cov_inv @ diff.T)

# Example: Fitting Gaussian distribution using Maximum Likelihood Estimation
f1_intact = extract_frequencies('intact', 'mode_1')
mu, sigma = stats.norm.fit(f1_intact)  # This performs MLE internally
plt.hist(f1_intact, bins=15, density=True, alpha=0.6, label='Data')
x = np.linspace(f1_intact.min(), f1_intact.max(), 200)
pdf = stats.norm.pdf(x, mu, sigma)
plt.plot(x, pdf, 'r-', linewidth=2, label=f'Fitted: mu={mu:.3f}, sigma={sigma:.3f}')
plt.legend()
```

8 GETTING STARTED

☐ Download `phase2_modal_parameters.json` from the course portal
☐ Load the data and confirm you have 20 estimates for each condition
☐ Identify your assigned sensor group from the metadata or Phase 1 instructions
☐ Compute mean mode shapes from intact measurements to use as MAC references
☐ Test the starter code and verify MAC calculations with the mean reference
☐ Create organized figure directories before generating plots
☐ Start with Part A to visualize feature distributions and understand the data patterns
☐ Work through Parts B-E sequentially, as later sections build on earlier concepts
☐ Keep notes on your observations to support the discussion in Part F
☐ Export notebook to PDF and verify all outputs are visible before submission